What Makes Good Form Good? Complexity, Transformational Invariance, and Aesthetic Value in Visual Organization

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Data and code can be found here `https://osf.io/28fbv/`

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Abstract

Visual patterns can be judged as simple, complex, good, symmetric, regular, or aesthetically pleasing. However, these judgments need not reflect the same underlying property. The present study deploys several objective measures of visual form, and views them as operational proxies for competing theories of perception. Observers rated a complete set of 3×3 binary patterns on perceived complexity, figural goodness, and aesthetic value. For each pattern, we computed local-statistical measures, algorithmic and compression-based measures, and Gestalt/group-theoretic measures based on reflection and rotation. The results revealed a clear dissociation between figural goodness and complexity. Subjective complexity was best captured by local-statistical and compression-like measures, whereas figural goodness was best accounted for by group-theoretic organization. Garner's rotation-and-reflection (R&R) set size strongly predicted figural goodness, even when algorithmic and compression-based measures were controlled for. Aesthetic value showed a partially similar pattern. These findings show that figural goodness is not simply low complexity. It is a judgment of organized form, that is best explained by transformational structure rather than by generic compressibility.

Keywords: figural goodness, visual complexity, algorithmic complexity, symmetry, Gestalt organization, aesthetic preference

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Over more than a century, the Gestalt tradition in psychology (for a review, see Wagemans, Elder, et al., 2012; Wagemans, Feldman, et al., 2012) has exerted a tremendous impact on our understanding of figural goodness, aesthetics, and complexity. Gestaltists such as Wertheimer (Wertheimer, 1912), Koffka (Koffka, 1935), and Köhler (Kohler, 1929) asked why some visual patterns are experienced as simple, orderly, and aesthetically pleasing, whereas other patterns appear disorganized, irregular, and random. These researchers argued that perception is mediated by highly stable, coherent, and organized structures. Good figures may have the same elements as less good figures, but they are perceived as more simple, balanced, and memorable. The Gestalt concept of figural goodness has been explained by such objective concepts as symmetry, regularity, closure, and Prägnanz (Koffka, 1935), and has exerted great influence on psychological thinking. But it has always been criticized for being difficult to quantify (Wagemans, Feldman, et al., 2012). Thus, psychologists have long searched for quantitative descriptions that will gauge the essence of good figures and their behavioral consequences (Garner, 1974; Palmer, 1991; Pomerantz & Kubovy, 1986). The present study therefore tests whether figural goodness is merely another name for low complexity, or whether it reflects a distinct principle of visual organization that is best captured by group-theoretic structure.

Information Theory and Figural Goodness

The first attempts to define figural goodness in quantitative terms were made in the wake of the cognitive revolution (Neisser, 1967). Information theory and information-processing frameworks were central to the cognitive revolution. Psychologists harnessed what was then Shannon's (Shannon, 1948) new invention of information theory—a mathematical theory of communication. Psychologists used measures of entropy, uncertainty, randomness, and redundancy as objective definitions, replacing older qualitative Gestalt concepts (Attneave, 1954, 1959; Falk, 1975; Falk & Konold, 1997;

Hochberg & McAlister, 1953). Good figures were defined as more redundant and therefore as transmitting less information than irregular figures (Garner, 1962, 1970; Garner & Clement, 1963). Garner's central insight was that a pattern's goodness can be understood from the size of the set of alternatives generated by rotating and reflecting it. Good patterns have few alternatives because they remain invariant across many transformations, whereas poor patterns generate many alternatives (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). Empirically, Garner and Clement (1963) showed that the size of the inferred set was strongly related to observers' figural-goodness judgments. The work of Garner (Garner, 1962, 1970; Garner & Clement, 1963), Attneave (Attneave, 1954, 1959), Hochberg (Hochberg & McAlister, 1953), and Falk (Falk, 1975; Falk & Konold, 1997) has been particularly influential in demonstrating how objective measures of information are related to subjective aspects of perception and visual organization.

Algorithmic Complexity and Visual Organization

Later developments in algorithmic information theory (Chaitin, 1969; Kolmogorov, 1965; Li & Vitányi, 2008; Solomonoff, 1964) have led cognitive researchers to offer a fundamentally different perspective on visual complexity and perceptual organization. This perspective is also consistent with Simon's (Simon, 1972) broader view that complex systems can often be understood in terms of their organization, structure, and generative description. *Algorithmic complexity* asks how difficult an object is to generate. Kolmogorov (Kolmogorov, 1965) offered a measure now called after his name. A pattern or object that requires a longer computer program to produce is considered more complex. The length of this program is *Kolmogorov complexity* (Li & Vitányi, 2008). This measure is uncomputable, but several practical approximations have been proposed. These include compression-based measures such as *Lempel–Ziv complexity* (Lempel & Ziv, 1976; Sheinwald, Lempel, & Ziv, 1990; Ziv & Lempel, 1977), the *Effort-to-Compress* algorithm (Nagaraj & Balasubramanian, 2017; Nagaraj, Balasubramanian, & Dey, 2013), and, more recently, the *Coding Theorem Method* based on algorithmic probability (Zenil, Soler-Toscano, Dingle, & Louis, 2015). An earlier approach to structural complexity was

developed by Leeuwenberg (Leeuwenberg, 1969), who proposed that the complexity of perceptual patterns could be quantified through their structural descriptions. This approach was subsequently elaborated within Structural Information Theory (Leeuwenberg & van der Helm, 2015) and the holographic model of figural goodness (van der Helm, 2014; van der Helm & Leeuwenberg, 1996).

Mental Programs and Symbolic Languages of Thought

A parallel program of research concerns mental programs and symbolic languages of thought. Dehaene and colleagues (Dehaene, Al Roumi, Lakretz, Planton, & Sablé-Meyer, 2022) have postulated that humans can encode structure in several domain-specific internal languages. According to this approach, people use dedicated primitives and compositional rules that are domain specific, such as spatial sequences, auditory sequences, mathematics, and geometric shapes (Amalric et al., 2017; Planton et al., 2021; Sablé-Meyer, Ellis, Tenenbaum, & Dehaene, 2022; Sablé-Meyer et al., 2021). The relevant measure is not description length in an arbitrary code, but minimum description length in a psychologically plausible language of representation. Spatial sequences, for example, can be compressed through primitives such as symmetry, rotation, and repetition (Amalric et al., 2017). Binary auditory sequences can be compressed through recursive alternation and repetition rules (Planton et al., 2021). Static geometric shapes seem to recruit symbolic geometric primitives and program-like descriptions (Sablé-Meyer et al., 2022, 2021). The mental-program perspective gives currency to our present approach, which contrasts between generic algorithmic or compression-based measures (Lempel & Ziv, 1976; Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013; Zenil et al., 2015; Ziv & Lempel, 1977) and group-theoretic measures (Garner, 1962, 1966, 1970, 1974; Garner & Clement, 1963; Garner & Gottwald, 1968; Palmer, 1991). Here, we employ group-theoretic measures that can be considered as a type of symbolic languages of thought. This is because they provide a psychologically plausible language for visual form. They can capture symmetry, rotation, reflection, and invariance, and are not limited to generic compressibility.

From Garner's Dot Patterns to Zenil's Binary Arrays

The connection between these approaches is especially important for the present study. This is because Garner's group-theoretic account of figural goodness was developed with simple dot patterns arranged in small matrices (Garner, 1970; Garner & Clement, 1963). More recent algorithmic-complexity approaches provide comparable objective measures for simple two-dimensional binary arrays. Particularly relevant is the work of Zenil and colleagues (Zenil et al., 2015). These researchers estimated the algorithmic complexity of small two-dimensional binary arrays. To do so, these researchers have used two-dimensional Turing machines and Levin's Coding Theorem (Levin, 1974). The theorem links algorithmic probability and complexity. According to this theory, objects generated by many short programs have high algorithmic probability and low estimated Kolmogorov complexity. In contrast, rarely generated objects have higher estimated complexity. Zenil et al.'s implementation applies this logic by running large classes of small two-dimensional Turing machines and estimating how frequently each binary array is produced. The relevant measure is $K_{m,2D}$. It should be noted that this measure differs from standard compression approaches. This is because it estimates complexity from a generative distribution over possible outputs. The approach offers a marked advantage over conventional compression algorithms, because it is more stable for short strings and small arrays (Delahaye & Zenil, 2012; Zenil et al., 2015). Zenil et al. (Zenil et al., 2015) provided in their paper a full catalogue of two-dimensional Kolmogorov complexity for patterns of 3×3 binary patches, which we use here. These patterns are shown in Figure 1. They are ordered from lowest to highest estimated $K_{m,2D}$ complexity.

The Psychological Validity of Objective Complexity Measures

We ask whether formal measures of algorithmic complexity, such as those provided by Zenil et al. (2015) and related compression-based approaches (Lempel & Ziv, 1976; Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013; Ziv & Lempel, 1977), correspond to the subjective experience of visual organization and aesthetic appreciation (Algom & Fitousi, 2025; Fitousi & Algom, 2024; Garner, 1970; Garner & Clement, 1963; Palmer,

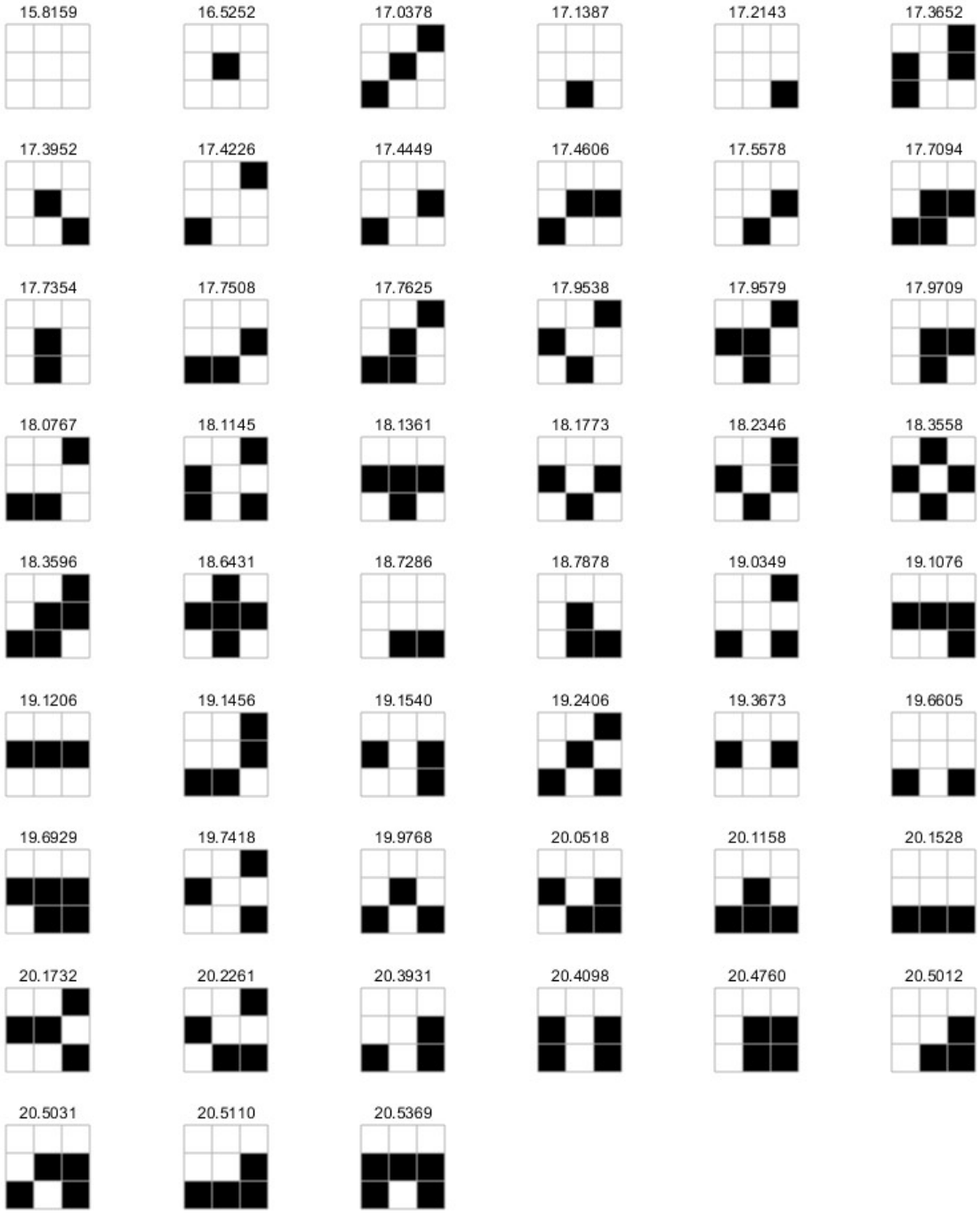


Figure 1. Set of 3×3 binary patterns as presented in Zenil et al. (2015). The figure is redrawn from Figure 6 of Zenil et al. (2015). The Figure presents the patterns from lowest to greatest estimated Kolmogorov complexity. Values above each patch denote the corresponding $K_{m,2D}$ estimate. The original article is open access and distributed under the Creative Commons Attribution 4.0 license.

1991). Their mathematical basis does not guarantee psychological validity: a pattern can have high algorithmic complexity but still be pleasing to the eye and elicit high figural goodness, while the reverse can also hold true. An early demonstration of the need to establish the psychological validity of objective measures was provided by Attneave (1957). Attneave related observers' complexity ratings of 72 nonrepresentational shapes to six measurable physical properties. He found that the number of contour turns was the strongest predictor of judged complexity.

Complexity, Figural Goodness, and Aesthetic Value

The relation between complexity, figural goodness, and aesthetic value is not straightforward (Van Geert, Bossens, & Wagemans, 2023; Van Geert & Wagemans, 2020, 2021). In classical aesthetics, pleasing form has often been linked to the balance between order and complexity (Birkhoff, 1933). A related idea in empirical aesthetics is that preference may follow an inverted-U relation with complexity: very simple patterns may be too impoverished, very complex patterns may be too irregular, and moderately complex patterns may be most aesthetically pleasing (Berlyne, 1971). This conjecture is valuable to the present study, because it entails that figural goodness and aesthetic value are not necessarily the same thing and therefore may be associated with different characteristics. For example, figural goodness may be related to symmetry, redundancy and transformational invariance, whereas aesthetic value may be related to interest, variety, novelty, and intermediate complexity. In the present work we also attempt to dissociate the two.

Competing Objective Measures of Structure

In recent work, we (Algom & Fitousi, 2025; Fitousi & Algom, 2024) have distinguished between objective measure of visual structure and subjective judgments of visual organization. The former are computed directly from the stimulus, whereas the latter are collected from observers. We pursue the same distinction here. Our objective measures come from various theoretical traditions. From information-theory we take local-statistical measures, such as entropy, transitions and density. From algorithmic

information theory we deploy algorithmic-complexity measures. These measures define complexity in terms of description length or generative probability (Chaitin, 1969; Kolmogorov, 1965; Levin, 1974; Li & Vitányi, 2008; Solomonoff, 1964). Zenil's two-dimensional complexity measure follows this tradition by using the Coding Theorem Method to estimate the algorithmic complexity of small two-dimensional binary arrays (Delahaye & Zenil, 2012; Zenil et al., 2015). Other measures, such as Lempel–Ziv complexity, originated in information theory and data compression and quantify sequential redundancy (Lempel & Ziv, 1976; Sheinwald et al., 1990; Ziv & Lempel, 1977).

Compression-like measures, such as the Effort-to-Compress algorithm, estimate the computational effort required to recursively simplify a symbolic sequence (Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013). Palindrome or SubSym measures quantify the prevalence of repeated symmetric substructures and can be extended from one-dimensional strings to two-dimensional images (Algom & Fitousi, 2025; Fitousi & Algom, 2024; Nagaraj & Balasubramanian, 2017; Toussaint, Onea, & Vuong, 2015).

Finally, we deploy symmetry (Weyl, 1952), and group-theoretic measures (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). These gauge invariance under reflection and rotation rather than algorithmic description length. As such, they provide different theoretical conception of complexity, goodness and organization, and therefore we do not expect them to yield identical psychological predictions. It is important to note that we use the term symmetry in this paper in a broader sense than mirror symmetry (Bertamini, Silvanto, Norcia, Makin, & Wagemans, 2018; Cattaneo, 2017; Palmer & Hemenway, 1978; Royer, 1981; Treder, 2010; Wagemans, 1997). Our usage employs the mathematical tradition (Carter, 2009; Weyl, 1952), which defines symmetry more generally as invariance under a group of transformations. This broader meaning of symmetry was harnessed in Garner's and Palmer's work (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). They argued that a pattern can be symmetric or good because it remains invariant under reflection, rotation, or combinations of transformations, not only because it has a single mirror axis. In the present work, we show empirically that our group-theoretic symmetry cannot be reduced to mirror

symmetry. The former notion has an explanatory power that goes far beyond the latter.

The Present Study

As can be noted from our introduction, many objective measures of complexity have been developed. However, only few studies have compared them directly using the same set of stimuli and participants (Algom & Fitousi, 2025; Fitousi & Algom, 2024). It is still unclear whether these measures gauge a common theoretical notion of visual complexity or rather they capture distinct structural properties. The literature (Berlyne, 1971; Birkhoff, 1933; Garner, 1970; Garner & Clement, 1963; Palmer, 1991)often lumps those theoretical measures together, while it is important to dissociate them. It is also not clear how these objective measures relate to subjective measures of figural goodness, aesthetic value, and complexity.

In the present study, we used the complete set of 51 non-equivalent 3×3 binary patterns introduced by Zenil et al. (2015). Participants rated each pattern on perceived complexity, figural goodness, and aesthetic value. The goal was to compare entire families of objective measures and relate them to subjective measures. For each pattern, we computed the objective measures. These were Lempel–Ziv complexity (Lempel & Ziv, 1976; Sheinwald et al., 1990; Ziv & Lempel, 1977), Effort-to-Compress (Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013), one- and two-dimensional SubSym complexity (Nagaraj & Balasubramanian, 2017; Toussaint et al., 2015), entropy and local statistical measures (Attneave, 1954; Hochberg & McAlister, 1953; Shannon, 1948), and group-theoretic symmetry measures (Garner, 1970; Garner & Clement, 1963; Palmer, 1991).

Objective Measures as Theoretical Proxies

We treat each family of measures as an operational proxy for a different theoretical account of visual organization (Algom & Fitousi, 2025; Fitousi & Algom, 2024). This framing is consistent with structural-information, algorithmic-complexity, and language-of-thought approaches, in which formal descriptions are psychologically informative only insofar as they approximate representational operations available to the

observer (Dehaene et al., 2022; Kolmogorov, 1965; Leeuwenberg, 1971; Li & Vitányi, 2008). The formal definitions of the measures, including the equations used to compute them, are provided in the Method section under Objective Measures. Our aim is to elucidate the theoretical meaning of each family of measures.

The first family of measures views patterns as collections of local elements. This family consists of density, entropy, transitions, run counts, and connectedness measures (Attneave, 1954; Hochberg & McAlister, 1953; Shannon, 1948). These measures capture the proportion of filled cells, cell-value uncertainty, local discontinuity, and fragmentation. Thus, we use them as a local-statistical proxy for perceived complexity.

The second family treats patterns as objects that can vary in description length, compressibility, or recoding difficulty. This family consists of algorithmic and compression-based measures, such as Zenil's $K_{m,2D}$, Lempel–Ziv complexity, Effort-to-Compress, and SubSym (Lempel & Ziv, 1976; Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013; Toussaint et al., 2015; Zenil et al., 2015; Ziv & Lempel, 1977). We use these measures as a proxy for a type of theoretical account that argues that visual organization depends on description length, algorithmic probability, or the availability of compact recodings (Dehaene et al., 2022).

The third family of indices that we employ is associated with the Gestalt tradition. These measures provide a proxy for the idea that good forms are organized as global wholes (Algom & Fitousi, 2016; Fitousi, 2013, 2015). These wholes remain stable under transformation (Garner & Clement, 1963). This family consists of group-theoretic measures such as Garner's rotation-and-reflection (R &R) set size and Palmer's symmetry-group size (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). It is important to note that we view symmetry, which is one of the most celebrated aspects of Gestalt, from a broader mathematical vantage point. Here symmetry is defined as invariance under a group of transformations (Carter, 2009; Weyl, 1952). In the present context, the relevant transformations are the rotations and reflections of the square. Table 1 summarizes the measure families, their representative measures, and their theoretical interpretation.

Table 1
Theoretical proxies for accounts of perceptual organization

Theoretical proxy	**Representative measures**	**Computational or perceptual assumption**	**Predicted psychological alignment**
Local-statistical proxy	Density; entropy; transition count; connected components; row-run and column-run counts	Observers summarize the proportion of filled cells, discontinuity, balance, and fragmentation without explicitly representing the global transformation structure of the whole pattern.	Primarily perceived complexity.
Algorithmic proxy	Zenil $K_{m,2D}$; Lempel–Ziv complexity; Effort-to-Compress; one-dimensional and two-dimensional SubSym indices	Observers are sensitive to description length, recoding difficulty, algorithmic probability, or the presence of compressible substructure.	Primarily perceived complexity, with possible weaker effects on figural goodness or aesthetic value.
Gestalt/group-theoretic proxy	Garner rotation-and-reflection set size; Palmer group size	Observers are sensitive to global organization under the rotations and reflections of the square. Good forms generate fewer distinct variants and are preserved by more transformations.	Primarily figural goodness, with converging effects on aesthetic value.

Note. The table summarizes the theoretical proxies used in the present study. The objective measures are not treated as isolated descriptors, but as operational proxies for competing accounts of perceptual organization. Formal definitions and equations for the measures appear in the Method section.

These three families imply different psychological alignments. Local-statistical measures should be most closely related to perceived complexity, because they quantify the proportion of filled cells, cell-value uncertainty, and local discontinuity (Attneave, 1954; Hochberg & McAlister, 1953; Shannon, 1948). Algorithmic and compression-based measures should predict judgments to the extent that observers rely on description length or recoding difficulty (Dehaene et al., 2022; Kolmogorov, 1965; Li & Vitányi, 2008; Zenil et al., 2015). Group-theoretic measures should predict judgments of figural goodness, if observers consider aspects of the whole figure, not just the elements that compose it. In the present effort, we ask three related questions using the same stimuli set. We ask

which family of objective measures best predicts: (a) subjective complexity, (b) figural goodness, and (c) aesthetic value. We expect that group-theoretic measure would exhibit the strongest link to subjective figural goodness. This is because they capture global transformational organization (Garner, 1970; Garner & Clement, 1963; Palmer, 1991).

Theoretical Predictions

The family-based framework leads to three competing predictions. The first prediction follows from local-statistical and information-theoretic accounts of visual form (Attneave, 1954; Garner, 1962; Hochberg & McAlister, 1953; Shannon, 1948). If observers base their judgments on local irregularity, then measures such as entropy, transitions, runs, and connected components should dominate the predictions. According to this account, figural goodness should decrease as local discontinuity and fragmentation increase.

The second prediction follows from algorithmic information theory and related compression-based approaches (Chaitin, 1969; Kolmogorov, 1965; Lempel & Ziv, 1976; Li & Vitányi, 2008; Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013; Solomonoff, 1964; Zenil et al., 2015; Ziv & Lempel, 1977). If subjective organization is determined by description length, one would expect the algorithmic family to provide the strongest account of all three judgment dimensions. In this case, patterns with low estimated algorithmic complexity, high compressibility, and large symmetric substructure should be judged: (a) as less complex, (b) better in term of figural-goodness, and (c) more aesthetically pleasing.

The third prediction follows from the Gestalt tradition and from Garner's and Palmer's group-theoretic accounts of figural goodness (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). If figural goodness is an aspect of global organization, then symmetry and transformation-invariance should be the strongest predictors of goodness. This prediction allows an important dissociation: a pattern may be locally complex and even algorithmically non-trivial, but still be perceived as a good figure if it has salient global symmetry.

Method

Participants

One hundred twenty two participants (N=122) were recruited from the Ariel community. Their age ranged from 19 to 74 years ($M = 41.36$, $SD = 13.97$). All participants provided informed consent before beginning the experiment. The study was approved by the Ethical Committee of Ariel University.

Stimuli

The stimuli were the complete set of 51 non-equivalent 3×3 binary patterns introduced by Zenil and colleagues (Zenil et al., 2015). Each stimulus consisted of a small binary array in which each cell was either filled or empty (see Figure 1) . This set is especially useful for the present purpose because it permits a direct comparison between the small-matrix stimuli used in the Garner tradition (Garner, 1970; Garner & Clement, 1963) and the two-dimensional binary arrays for which Zenil et al. provided algorithmic-complexity estimates (Zenil et al., 2015).

Design and Procedure

The study used a stimulus-level rating design. It was designed in Qualtrics and administered online through a distributed survey link. Participants viewed the complete set of 51 non-equivalent 3×3 binary patterns and evaluated each pattern on three subjective dimensions: perceived complexity, figural goodness, and aesthetic value. Each of the 51 patterns was presented three times to each participant: once for a perceived-complexity judgment, once for a figural-goodness judgment, and once for an aesthetic-value judgment. Thus, each participant was asked to complete 153 pattern-rating trials. The order of the three tasks was randomized across participants. The order of the patterns was also randomized across participants and tasks.

For each pattern, we averaged the ratings across participants to obtain three mean psychological judgments: subjective complexity, figural goodness, and aesthetic value. We also computed various objective measures for each pattern, which we outline in the

subsequent section. We tested the hypothesis that the different families of objective measures predict the same psychological judgments. The alternative hypothesis was that each family of measures gauges a different dimension of perceptual experience.

Subjective Judgments. For each pattern X_k, participants provided ratings in three separate blocks: (a) perceived complexity, (b) figural goodness, and (c) aesthetic value. In all judgment tasks, participants rated patterns on 7-point scales. For perceived complexity, the scale ranged from 1 ("not complex at all") to 7 ("very complex"). For figural goodness, the scale ranged from 1 ("bad figure") to 7 ("good figure"). Participants were also provided with a brief explanation describing good figures in terms of symmetry and harmony. For aesthetic value, the scale ranged from 1 ("low aesthetic value") to 7 ("high aesthetic value"), and participants judged how aesthetically pleasing each pattern was.

The main analyses were conducted at the stimulus level. We used the mean ratings across participants for each of the 51 patterns as dependent variables. This approach asks which objective properties of a pattern predict the average psychological impression that the pattern produces.

We analyzed each judgment dimension separately. The rationale is that each of these dimensions (i.e., complexity, figural goodness, and aesthetic value) are related but not identical psychological constructs (Berlyne, 1971; Birkhoff, 1933; Garner, 1970; Garner & Clement, 1963; Palmer, 1991). We assume that the three families of measures are not necessarily dependent. The empirical question is therefore not whether a measure is generally valid, but which psychological construct each family of measures best explains (Algom & Fitousi, 2025; Fitousi & Algom, 2024).

Objective Measures. Let a stimulus be represented by a binary matrix

$$X = \{x_{ij} : i, j = 1, \ldots, 3\}, \qquad x_{ij} \in \{0, 1\}, \tag{1}$$

where $x_{ij} = 1$ denotes a filled cell and $x_{ij} = 0$ denotes an empty cell. The total number of cells is $N = 9$. For measures that require a one-dimensional sequence, the matrix was

vectorized in row-major order,

$$s(X) = (x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}). \tag{2}$$

Other vectorizations are possible, and this issue is important because several compression-based measures are order dependent (Lempel & Ziv, 1976; Sheinwald et al., 1990; Ziv & Lempel, 1977). We therefore computed row-major versions of the one-dimensional compression measures and versions averaged across eight readings. The eight readings, denoted $s_1(X), \ldots, s_8(X)$, were obtained by applying the four rotations of the square and, for each rotation, its horizontal reflection, and then vectorizing the resulting matrix in row-major order. The row-major and eight-reading versions reported below correspond to the measures included in the analyses.

Local-Statistical Measures. The simplest descriptor is the density of filled cells,

$$D(X) = \frac{1}{N} \sum_{i=1}^{3} \sum_{j=1}^{3} x_{ij}. \tag{3}$$

Density is not a complexity measure by itself, but it is important because many information-theoretic and perceptual descriptions are affected by the proportion of the two element types (Attneave, 1954; Attneave & Arnoult, 1956; Shannon, 1948).

Shannon entropy was computed from the empirical probabilities of filled and empty cells (Attneave, 1954; Shannon, 1948). Let $p = D(X)$. The entropy of the pattern is

$$H(X) = -p \log_2 p - (1 - p) \log_2 (1 - p), \tag{4}$$

with the convention that $0 \log_2 0 = 0$. Entropy is maximal when filled and empty cells are equally frequent, and it decreases as the pattern becomes dominated by one cell type.

Local spatial variation was quantified by counting horizontal and vertical transitions between adjacent cells. This measure captures local discontinuity. It is consistent with early information-theoretic, which applied local-statistical approaches to

visual form (Attneave, 1954; Attneave & Arnoult, 1956; Julesz, 1962),

$$T(X) = \sum_{i=1}^{3}\sum_{j=1}^{2} |x_{ij} - x_{i,j+1}| + \sum_{i=1}^{2}\sum_{j=1}^{3} |x_{ij} - x_{i+1,j}|. \tag{5}$$

In a 3×3 grid there are twelve horizontal and vertical adjacencies. Consequently, $T(X)$ ranges from 0 to 12. A high transition count indicates frequent local changes, whereas a low count indicates local homogeneity.

We define a run as a maximal sequence of identical consecutive symbols (Lempel & Ziv, 1976). For example, the sequence 0011100 contains three runs: 00, 111, and 00. We counted the runs for each pattern separately along rows and columns. The row-run and column-run counts were defined as

$$R_{\text{row}}(X) = \sum_{i=1}^{3} \left[1 + \sum_{j=1}^{2} \mathbb{I}\{x_{ij} \neq x_{i,j+1}\} \right], \tag{6}$$

and

$$R_{\text{col}}(X) = \sum_{j=1}^{3} \left[1 + \sum_{i=1}^{2} \mathbb{I}\{x_{ij} \neq x_{i+1,j}\} \right]. \tag{7}$$

Each count can range from 3 (i.e., when every row or column is homogeneous), to 9, (i.e., when every row or column alternates at both adjacencies). These measures allow us to distinct between horizontal and vertical organization.

Connectedness was defined using four-neighbor adjacency. This is standard definition of connectivity for binary digital images (Rosenfeld & Pfaltz, 1966). It reflects the Gestalt idea that spatial grouping and coherence contribute to perceived organization (Koffka, 1935; Pomerantz & Kubovy, 1986). Let $C(X)$ be the number of connected components formed by filled cells. Importantly, two filled cells belong to the same component if they are linked through a sequence of horizontal or vertical adjacencies. Diagonal contact was not sufficient for connectivity, and components formed by empty cells were not included. Formally,

$$C(X) = \#\{\text{four-neighbor connected components of filled cells in } X\}. \tag{8}$$

Patterns with many disconnected filled regions receive higher values, whereas patterns in which the filled cells form a single coherent region receive lower values.

Algorithmic and Compression-Based Measures. Because exact Kolmogorov complexity is not computable in general, the analyses used computable algorithmic and compression-based measures (Chaitin, 1969; Kolmogorov, 1965; Li & Vitányi, 2008; Solomonoff, 1964; Zenil et al., 2015).

Lempel–Ziv complexity estimates the number of phrases required to parse a sequence (Lempel & Ziv, 1976; Sheinwald et al., 1990; Ziv & Lempel, 1977). Let $c_{LZ}^{(a)}(s)$ denote the phrase count produced by parsing algorithm a, where the analyses included both exhaustive and primitive parsing. For each algorithm, we computed a row-major value and an average across the eight readings,

$$LZ_{a,\mathrm{row}}(X) = c_{LZ}^{(a)}(s(X)), \qquad LZ_{a,\mathrm{mean8}}(X) = \frac{1}{8}\sum_{k=1}^{8} c_{LZ}^{(a)}(s_k(X)). \tag{9}$$

Higher values indicate that the readings require more phrases and are therefore less compressible. The principal Lempel–Ziv result reported in the text used primitive complexity averaged across the eight readings; the other variants were included in the family-level figures and robustness analysis.

Effort-to-Compress measures the number of recursive substitutions required to reduce a sequence to a constant sequence (Nagaraj & Balasubramanian, 2017; Nagaraj et al., 2013). Let $\phi(s)$ denote one application of the non-sequential recursive pair-substitution operation, and let $q(s)$ be the smallest integer for which $\phi^{q(s)}(s)$ is constant. We computed the row-major value and the average across the eight readings,

$$ETC_{\mathrm{row}}(X) = q(s(X)), \qquad ETC_{\mathrm{mean8}}(X) = \frac{1}{8}\sum_{k=1}^{8} q(s_k(X)). \tag{10}$$

A small value means that the readings can be simplified quickly, whereas a large value means that more recursive operations are required.

The Coding Theorem Method estimates algorithmic complexity from algorithmic probability (Delahaye & Zenil, 2012; Levin, 1974; Zenil et al., 2015). Let $AP(X)$ denote

the probability that a randomly sampled program, or an enumerated computational process from the relevant machine class, generates X. Zenil et al.'s two-dimensional estimate was

$$K_{m,2D}(X) = -\log_2 AP(X) + c, \tag{11}$$

where c is an additive constant that does not affect the ordering of patterns.

SubSym measures quantify the amount of symmetric substructure in a sequence or image (Nagaraj & Balasubramanian, 2017; Toussaint et al., 2015). For the one-dimensional version, let $\mathcal{U}(s)$ be the set of all contiguous substrings of $s(X)$ with length at least 2. In accordance with the implementation used in the analyses, each symmetric substring was weighted by its length. The one-dimensional SubSym score is

$$SS_{1D}(X) = \sum_{u \in \mathcal{U}(s)} |u| \, \mathbb{I}\{u = \mathrm{rev}(u)\}, \tag{12}$$

where $\mathrm{rev}(u)$ is the reversed substring and $\mathbb{I}\{\cdot\}$ is the indicator function. To express this as a complexity measure rather than a regularity score, the normalized SubSym complexity is

$$SubSym_{1D}(X) = 1 - \frac{SS_{1D}(X)}{SS_{1D}^{\max}}, \tag{13}$$

where the normalizing constant for a sequence of length n was

$$SS_{1D}^{\max}(n) = \sum_{\ell=2}^{n} \ell(n - \ell + 1). \tag{14}$$

Here, n denotes the length of the sequence, and ℓ denotes the length of a contiguous substring, ranging from 2 to n. This is the score obtained when every contiguous substring of length at least 2 is symmetric.

For the one-dimensional measure, the analyses included the raw and normalized scores for the row-major reading and their averages across the eight readings. Thus, for either the raw or normalized function f,

$$f_{\mathrm{row}}(X) = f(s(X)), \qquad f_{\mathrm{mean8}}(X) = \frac{1}{8} \sum_{k=1}^{8} f(s_k(X)). \tag{15}$$

Here, $s(X)$ denotes the row-major reading of pattern X, whereas $s_k(X)$ denotes its kth reading among the eight possible readings, with $k = 1, \ldots, 8$.

The two-dimensional version was computed by applying the one-dimensional SubSym score to the three rows, three columns, and all diagonals of length at least two in both diagonal orientations. Let $\mathcal{L}(X)$ denote this collection of one-dimensional lines. The raw two-dimensional score was

$$SS_{2D}(X) = \sum_{u \in \mathcal{L}(X)} SS_{1D}(u). \tag{16}$$

The corresponding normalized complexity index was

$$SubSym_{2D}(X) = 1 - \frac{SS_{2D}(X)}{SS_{2D}^{\max}}. \tag{17}$$

In the implementation, $SS_{2D}^{\max}$ was obtained by applying the same two-dimensional scoring procedure to an all-filled 3×3 matrix. The analyses included both the raw two-dimensional score and the normalized two-dimensional complexity index. The normalized index decreases when a pattern contains many symmetric substructures and increases when such substructures are rare.

Group-Theoretic Measures. The Gestalt measures were defined in the classical tradition of figural goodness, in which the goodness of a pattern is determined by the set of transformations under which the pattern is preserved. In contrast to statistical measures, these measures do not count elements, transitions, or connected components. In contrast to algorithmic measures, these measures do not estimate compressibility. Instead, they ask whether a given pattern belongs to a small and highly structured equivalence class under the rotations and reflections of the square. This approach follows Garner's R &R account of pattern goodness and Palmer's group-theoretic reformulation of the same idea (Garner, 1970; Garner & Clement, 1963; Palmer, 1991).

To define the Garner measure, start by letting D_4 denote the dihedral group of the square. This group contains the eight transformations that preserve the square lattice

(Palmer, 1991; Weyl, 1952):

$$D_4 = \{e, \rho_{90}, \rho_{180}, \rho_{270}, v, h, d, a\}, \tag{18}$$

where e is the identity transformation, ρ_{90}, ρ_{180}, and ρ_{270} are rotations by 90°, 180°, and 270°, and v, h, d, and a are reflections about the vertical, horizontal, main-diagonal, and anti-diagonal axes, respectively. For a pattern X, the transformed pattern obtained by applying $g \in D_4$ is denoted by $g(X)$.

Garner's Rotation-and-Reflection (R&R) Set Size. Garner's R &R measure is based on the number of distinct patterns generated by applying all rotations and reflections to a given pattern. For each pattern X, we define its transformation set, or orbit, as

$$\mathcal{O}(X) = \{g(X) : g \in D_4\}. \tag{19}$$

Garner's rotation-and-reflection set size is the number of distinct members of this set:

$$R(X) = |\mathcal{O}(X)| = |\{g(X) : g \in D_4\}|. \tag{20}$$

The logic of this measure is that good patterns generate few distinct alternatives under rotation and reflection (Garner, 1970; Garner & Clement, 1963). If many transformations produce the same pattern, the pattern is highly regular. If each transformation produces a different pattern, the pattern is relatively irregular. Thus, $R(X)$ is an inverse measure of figural goodness. Small values of $R(X)$ indicate high goodness, whereas large values indicate low goodness.

$R(X)$ ranges from 1 to 8. This is because the group D_4 contains eight transformations. Thus, a completely invariant pattern has $R(X) = 1$ because all rotations and reflections generate the same pattern. A pattern with no nontrivial rotational or reflectional invariance has $R(X) = 8$ since all its eight transformations generate distinct patterns.

Palmer's Symmetry-Group Measure. Palmer's group-theoretic formulation describes the same organizational property from the complementary perspective (Palmer,

1991). Instead of asking how many distinct patterns are generated by transformations, Palmer's measure asks how many transformations leave the pattern unchanged. For each pattern X, its symmetry group, or stabilizer, is defined as

$$\text{Stab}(X) = \{g \in D_4 : g(X) = X\}. \tag{21}$$

The Palmer symmetry-group size is then

$$P(X) = |\text{Stab}(X)| = \sum_{g \in D_4} \mathbf{1}\{g(X) = X\}, \tag{22}$$

where $\mathbf{1}\{g(X) = X\}$ equals 1 if transformation g leaves the pattern unchanged and 0 otherwise. Larger values of $P(X)$ indicate that the pattern is invariant under more transformations and is therefore more organized. For example, if only the identity transformation leaves the pattern unchanged, then $P(X) = 1$. But if all eight rotations and reflections leave the pattern unchanged, then $P(X) = 8$. Thus, Palmer's measure is a direct measure of transformational invariance: larger values indicate greater symmetry and greater figural goodness.

The relation between Garner's and Palmer's measures is given by the orbit-stabilizer theorem (Carter, 2009; Palmer, 1991; Weyl, 1952). Because D_4 has eight elements, $R(X)P(X) = |D_4| = 8$, and therefore $P(X) = 8/R(X)$. This means that Garner's and Palmer's measures are not independent predictors. They are two inverse descriptions of the same group-theoretic property. Garner's measure counts the number of distinct alternatives produced by transforming the pattern; Palmer's measure counts the number of transformations that preserve the pattern. In the present analyses, Garner's $R(X)$ was used as an inverse measure of goodness, whereas Palmer's $P(X)$ was used as a direct measure of goodness.

Figure 2 provides an illustration of the group-theoretic logic. It uses examples from the present stimulus set. We present this figure to clarify that Garner's measure is not: (a) a count of local elements and not (b) a conventional mirror-symmetry score. It is the size of the set of distinct alternatives generated by the rotations and reflections of the

square. Patterns with low values of $R(X)$ remain invariant under many transformations, whereas patterns with high values generate many distinct alternatives. This visualization follows Garner's original rotation-and-reflection analysis and later applications to figural goodness and complexity, but uses examples from the Zenil set rated in the present experiment (Algom & Fitousi, 2025; Fitousi & Algom, 2024; Garner, 1970; Garner & Clement, 1963).

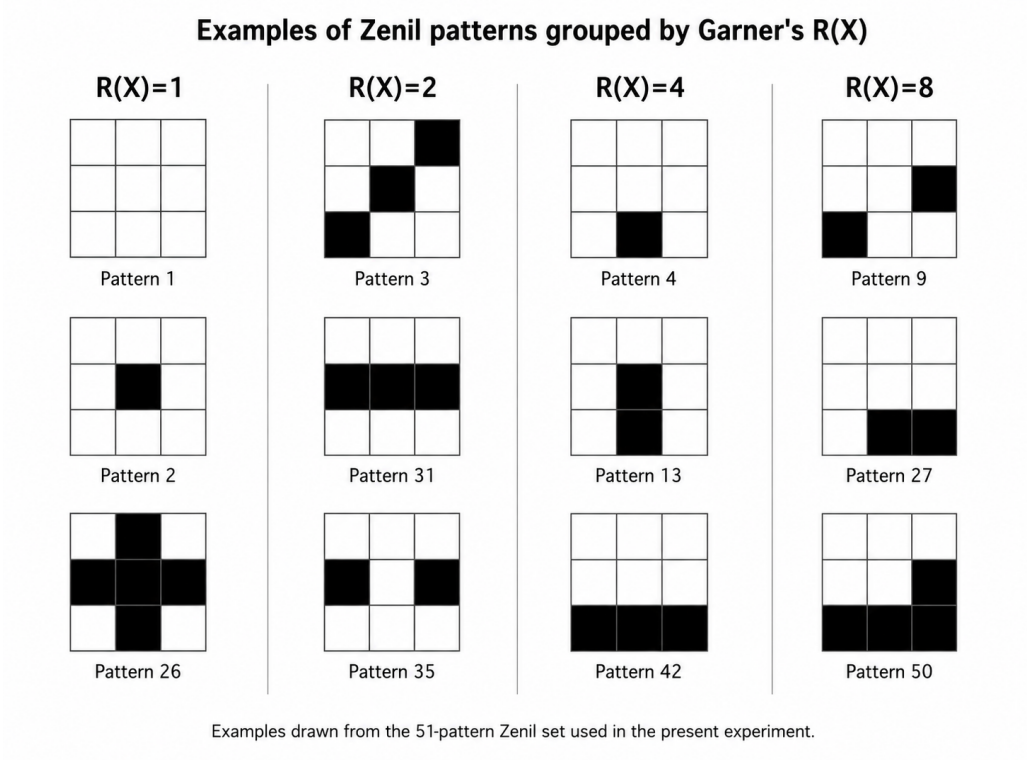


Figure 2. Examples of Zenil patterns grouped by Garner's rotation-and-reflection set size, $R(X)$. For each pattern, $R(X)$ is the number of distinct alternatives generated by applying the eight rotations and reflections of the square. Low values of $R(X)$ indicate greater transformational invariance, whereas high values indicate fewer preserved transformations and more distinct alternatives. The examples are drawn from the stimulus set used in the present study.

This family of measures is important for our psychological account. It implements the Gestalt idea that good figures are those that remain stable under transformations of the visual field (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). A good pattern is a pattern whose global organization is preserved by the natural transformation group of the square.

Analysis Strategy

The analyses were conducted at the stimulus level. We used the mean rating assigned to each pattern. Predictors and dependent variables were standardized before regressing or correlating them, so we can compare them across families. Importantly, the purpose of these analyses was to compare different theoretical approaches to perceptual organization. Each family of predictors was treated as an empirical representation of a different theoretical approach to explaining subjective judgments of visual patterns.

The local-statistical family served as a proxy for a parallel summary-statistic account, according to which observers judge patterns by pooling the proportion of filled cells, local discontinuity, and fragmentation (Attneave, 1954; Hochberg & McAlister, 1953; Shannon, 1948). The algorithmic and compression-based family served as a proxy for a description-length or recoding account, according to which patterns are experienced as simple or complex according to their compressibility (Kolmogorov, 1965; Lempel & Ziv, 1976; Li & Vitányi, 2008; Zenil et al., 2015; Ziv & Lempel, 1977). The group-theoretic family served as a proxy for a Gestalt account, according to which figural goodness depends on global invariance under rotations and reflections (Garner, 1970; Garner & Clement, 1963; Palmer, 1991).

Our analysis followed these four steps. First, we estimated the pairwise correlations between each objective measure and each subjective judgment. We used the observed correlations to determine which objective measures were related with subjective measures of complexity, figural goodness, and aesthetic value. Second, we compared the families of objective measures: (a) local-statistical measures, (b) algorithmic and compression-based measures, and (c) group-theoretic measures. We then performed local-statistical regression with density, entropy, transition count, and connected components. Row-run and column-run counts were retained in the pairwise analyses and figures but were omitted from this family regression because transition count is exactly equal to the sum of the row-run and column-run counts minus six. Third, we fit a combined model that included: (a) Garner's rotation-and-reflection set size, (b) normalized two-dimensional SubSym complexity, (c) mean-eight Effort-to-Compress, (d) mean-eight primitive

Lempel–Ziv complexity, and (e) Zenil's $K_{m,2D}$ estimate. In this model, we tested whether the group-theoretic proxy retained predictive value after controlling for algorithmic and compression-based predictors. Garner's R&R set size and Palmer's group size were not entered together in the main regression model because they are mathematically related. Across the 51 patterns, the two measures were strongly negatively correlated ($r = -.780$, $p < .001$). Fourth, we tested the intermediate complexity account (Berlyne, 1971) for each judgment dimension. Each model contained a standardized linear term and its squared term. Note that Garner's R&R set size is a variable with discrete values. Thus, correlation and regression lines with this variable were interpreted as indicating order rather than continuous relations.

Model fit was summarized by the ordinary coefficient of determination, $R^2 = 1 - \sum_k (y_k - \hat{y}_k)^2 / \sum_k (y_k - \bar{y})^2$, and by the adjusted coefficient of determination, $R^2_{adj} = 1 - (1 - R^2)(K - 1)/(K - p - 1)$, where K is the number of patterns and p is the number of predictors. To estimate out-of-sample performance, we also used leave-one-out cross-validation. For each pattern k, the model was refit after omitting that pattern, and the omitted rating was predicted from the fitted model. Cross-validated fit was summarized as $R^2_{LOOCV} = 1 - \sum_k (y_k - \hat{y}_{k,-k})^2 / \sum_k (y_k - \bar{y})^2$, together with $RMSE_{LOOCV} = \sqrt{K^{-1} \sum_k (y_k - \hat{y}_{k,-k})^2}$. These quantities are reported as model diagnostics; the theoretical interpretation rests on the family of perceptual operations that each predictor operationalizes.

Results

It can be constructive, before we turn to the statistical analyses, to obtain a first visual impression of the participants' judgments. Figure 3 presents the five patterns with the highest and lowest mean ratings for each of the three subjective judgments: figural goodness, aesthetic value, and subjective complexity. This figure can give the reader intuition regarding the three types of judgments. As can be noted, the patterns receiving the highest ratings for figural goodness and aesthetic value largely overlapped but were not exactly the same. A similar overlap can be observed among the lowest-rated patterns

for these two judgments, whereas the patterns receiving the lowest complexity judgments were largely different from both sets.

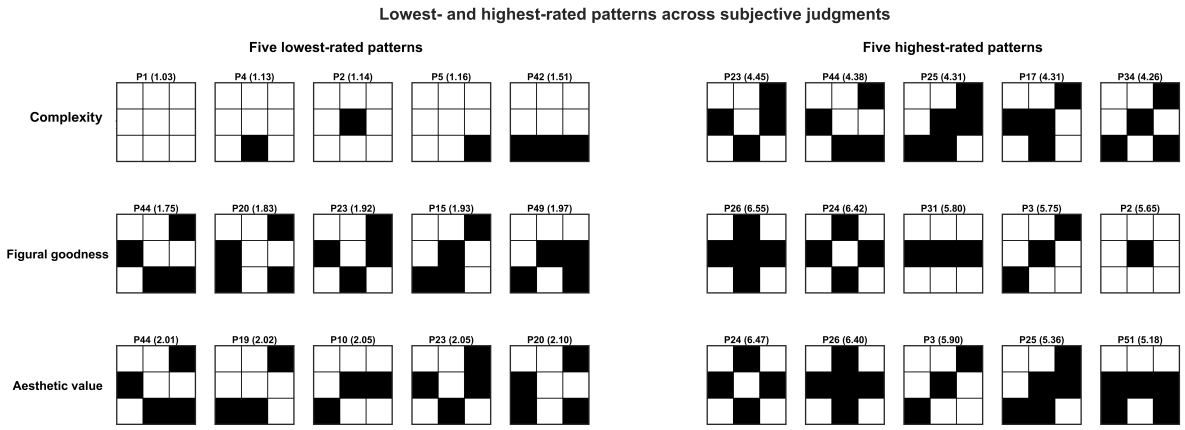


Figure 3. The five lowest-rated (left) and five highest-rated (right) patterns for each of the three subjective judgments: subjective complexity, figural goodness, and aesthetic value. Patterns are ranked according to their mean ratings across participants. The pattern identification number (P) and corresponding mean rating are displayed above each pattern. All judgments were made on 7-point scales.

A complementary visualization is shown in Figure 4. Here the rows show the five non-uniform patterns occupying the low-complexity or high-invariance extreme of representative objective measures from the group-theoretic, algorithmic/compression-based, and local-statistical families. Uniform patterns were omitted from this descriptive figure because the completely empty pattern constitutes a trivial minimum for most complexity measures and would therefore recur across the rows; it was retained in all statistical analyses. For Garner's rotation-and-reflection set size, the figure shows the five lowest-scoring non-uniform patterns, which have the greatest transformational invariance and hence the greatest predicted figural goodness. For Zenil complexity, Lempel–Ziv complexity, Effort-to-Compress, and entropy, the figure likewise shows the five lowest-scoring non-uniform patterns, which are the patterns identified as least complex by each measure. The purpose of this figure is to show visually whether different objective conceptions of simplicity and good form identify the same nontrivial patterns. The formal analyses below quantify these relations.

Before relating the judgments to the objective measures, we examined the pattern-level correlations among the three subjective dimensions. Subjective complexity

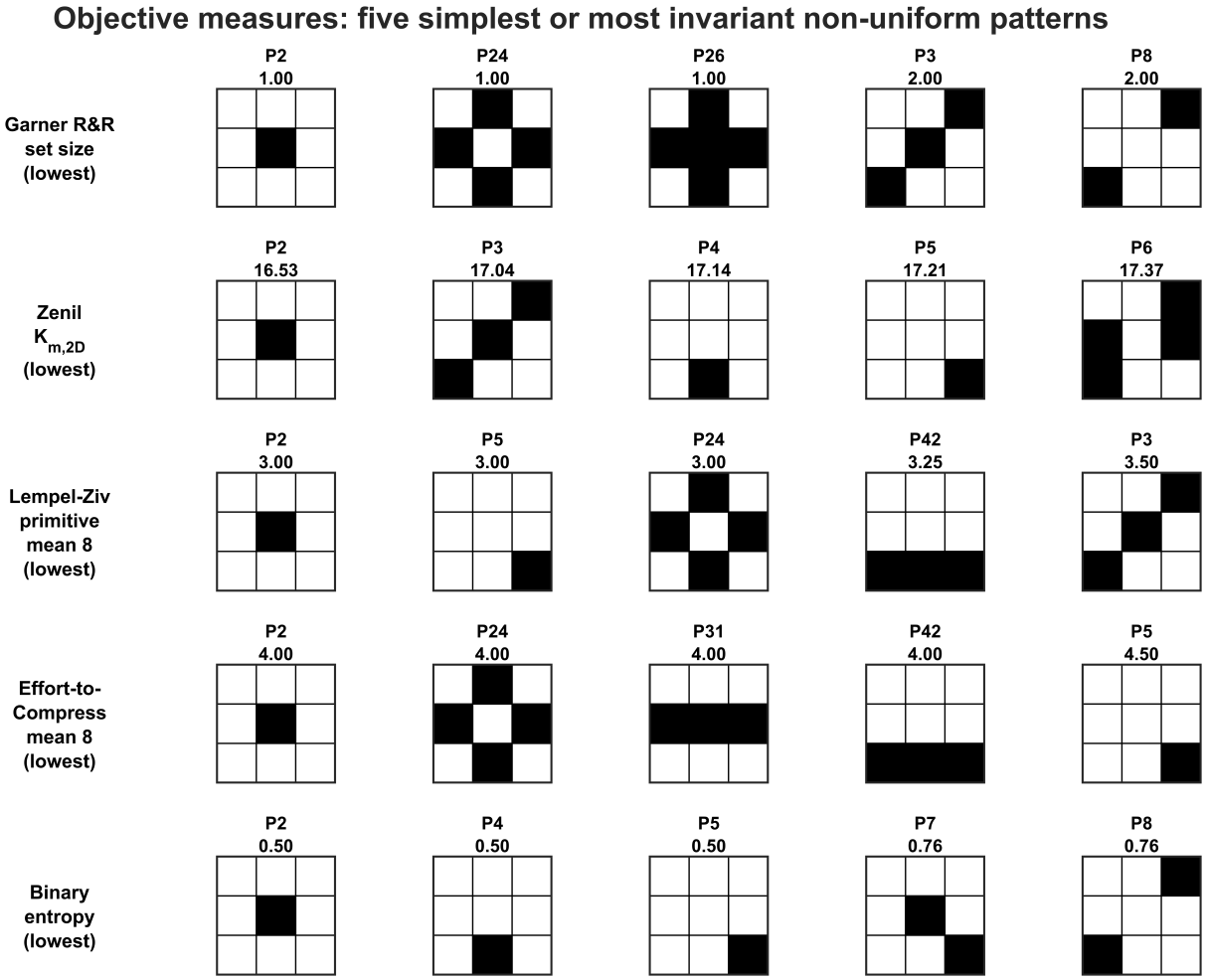


Figure 4. Patterns selected by representative objective measures. The figure shows the five simplest or most invariant non-uniform patterns according to each objective measure. The first row shows the five patterns with the lowest Garner rotation-and-reflection set size; these patterns have the greatest transformational invariance and hence the greatest predicted figural goodness. The remaining rows show the five patterns with the lowest Zenil two-dimensional algorithmic complexity, Lempel–Ziv primitive complexity averaged across readings, Effort-to-Compress averaged across readings, and binary cell-value entropy; these are the patterns identified as least complex by each measure. Each cell gives the pattern identifier (i.e.. P5), and the corresponding objective value.

was negatively related to figural goodness, $r(49) = -.49$, $p < .001$, and to aesthetic value, $r(49) = -.32$, $p = .022$. Figural goodness and aesthetic value were very strongly related, $r(49) = .96$, $p < .001$, sharing approximately 92% of their pattern-level variance. Thus, complexity was related to, but clearly dissociable from, the other judgments, whereas figural goodness and aesthetic value provided highly convergent evaluations of this stimulus set.

We next examined the relationships between the objective measures and the three subjective judgments. Table 2 provides an overview of selected pairwise correlations across the three theoretical families. The following sections examine these relationships separately for subjective complexity, figural goodness, and aesthetic value.

Table 2
Selected pairwise correlations between theoretical proxies and subjective judgments

Measure	SC r	SC R^2	FG r	FG R^2	AV r	AV R^2
Garner rotation-and-reflection set size	.48	.23	−.84	.71	−.80	.64
Palmer group size	−.41	.17	.69	.48	.63	.40
SubSym2D normalized	.78	.61	−.35	.12	−.20	.04
LZ primitive mean-eight	.68	.47	−.30	.09	−.19	.04
ETC mean-eight	.47	.22	−.29	.08	−.16	.03
Density	.75	.56	−.06	.00	.09	.01
Entropy	.68	.46	−.15	.02	.01	.00
Transition count	.68	.47	.00	.00	.17	.03
Zenil $K_{m,2D}$ complexity	.26	.07	−.04	.00	−.03	.00

Note. SC = subjective complexity; FG = figural goodness; AV = aesthetic value. Values are Pearson correlations and squared correlations. The table is organized as a comparison among theoretical proxies rather than as a search for a single universal complexity measure.

Subjective complexity is aligned with local-statistical and compression proxies

We first examined whether subjective complexity was best characterized by local statistics, algorithmic compressibility, or group-theoretic organization. The results showed that subjective complexity was not dominated by the same proxy that predicted figural goodness. Instead, subjective complexity was strongly related to measures that capture the proportion of filled cells, irregularity, and compressibility. Density was strongly related to subjective complexity, $r(49) = .75$, $p < .001$, $R^2 = .56$, as were transition count, $r(49) = .68$, $p < .001$, $R^2 = .47$, entropy, $r(49) = .68$, $p < .001$,

$R^2 = .46$, and Lempel–Ziv primitive complexity averaged across readings, $r(49) = .68$, $p < .001$, $R^2 = .47$. Normalized two-dimensional SubSym complexity was also strongly related to subjective complexity, $r(49) = .78$, $p < .001$, $R^2 = .61$.

This pattern supports the notion that subjective complexity is not a measure of global figural organization. It is more a measure of the quantity of filled visual material, irregularity, and compressibility. Figures 5, 6, and 7 give the corresponding family-level scatterplots. The local-statistical panels provide the clearest evidence for the summary-statistic proxy: patterns with greater density, more transitions, and more fragmentation tended to be judged as more complex. The algorithmic and SubSym panels show that compression-like proxies also tracked subjective complexity, although not all algorithmic measures were equally effective. Zenil et al.'s $K_{m,2D}$ estimate was only weakly related to subjective complexity, $r(49) = .26$, $p = .068$, $R^2 = .07$. Across the 51 patterns, $K_{m,2D}$ ranged from 15.82 to 20.54 ($M = 18.71$, $SD = 1.23$).

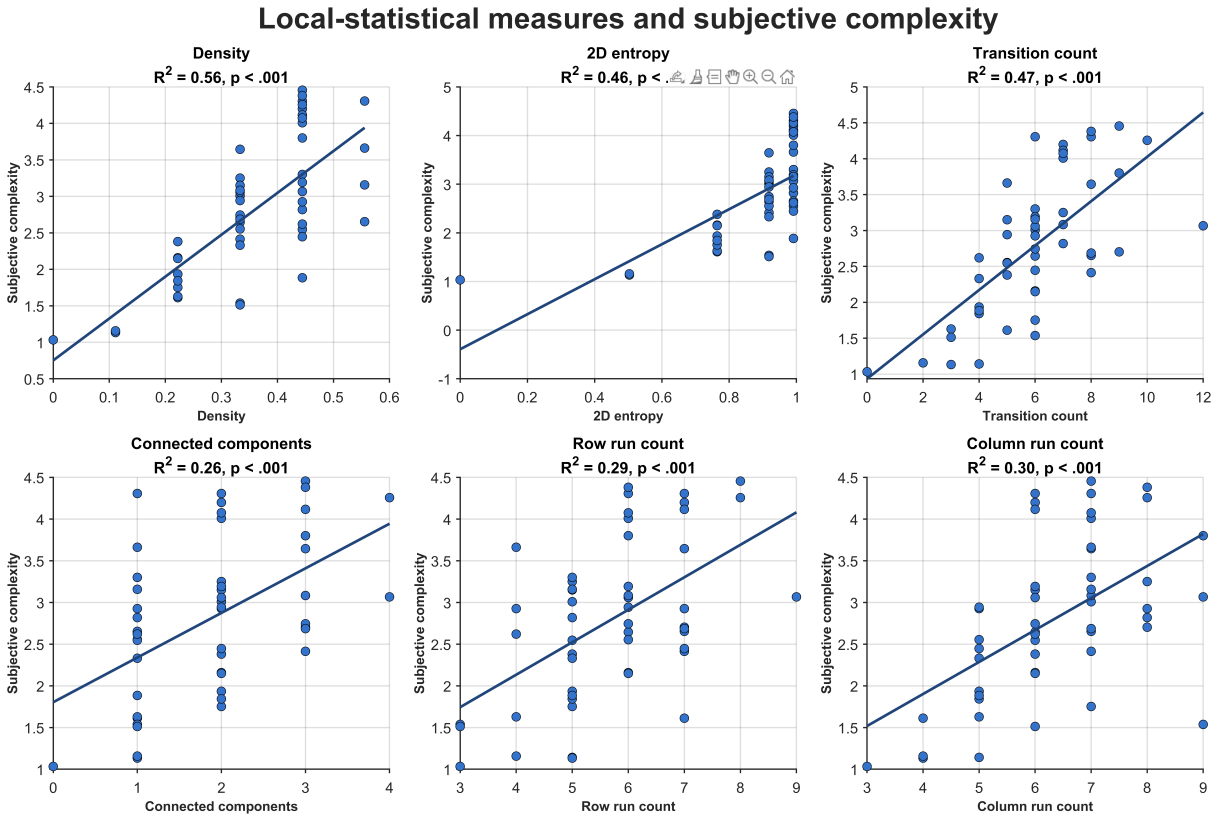


Figure 5. The local-statistical proxy for subjective complexity. The panels show density, entropy, transition count, connected components, row-run count, and column-run count. Each point represents one 3×3 binary pattern.

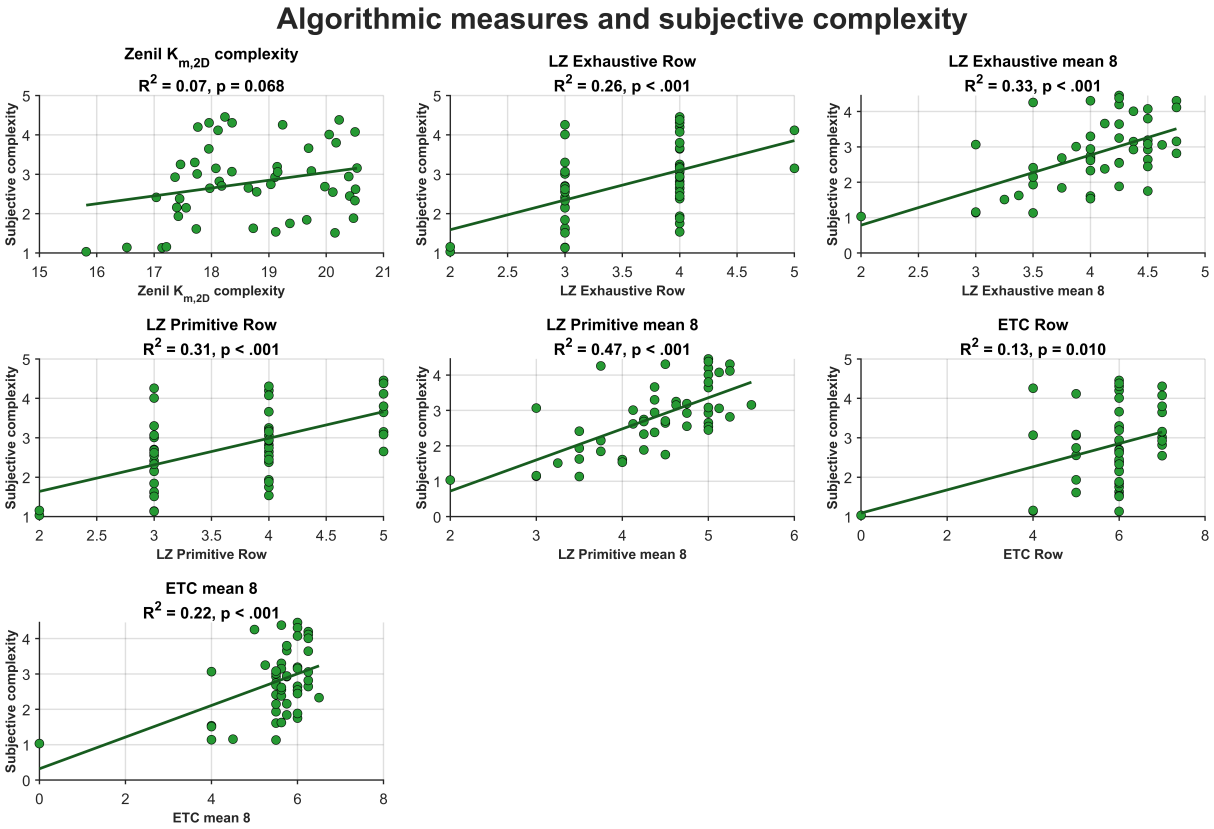


Figure 6. The algorithmic-compression proxy for subjective complexity. The panels show Zenil et al.'s $K_{m,2D}$ estimate, Lempel–Ziv complexity, and Effort-To-Compress indices. Each point represents one 3×3 binary pattern. Each panel includes the fitted linear trend and the corresponding R^2 and p value. These plots test the description-length account, according to which perceived complexity reflects compressibility or ease of recoding.

The Gestalt proxy predicts figural goodness

The main prediction of the Gestalt account was that figural goodness should be related to transformational organization. This prediction was strongly supported. Garner's rotation-and-reflection set size was the strongest single predictor of figural goodness, $r(49) = -.84$, $p < .001$, $R^2 = .71$. Thus, patterns with fewer R &R alternatives were judged as having greater figural goodness. Palmer's group-size measure also showed a strong relation with figural goodness, $r(49) = .69$, $p < .001$, $R^2 = .48$, indicating that patterns preserved by more transformations were judged as better forms. These relations are shown in Figure 8.

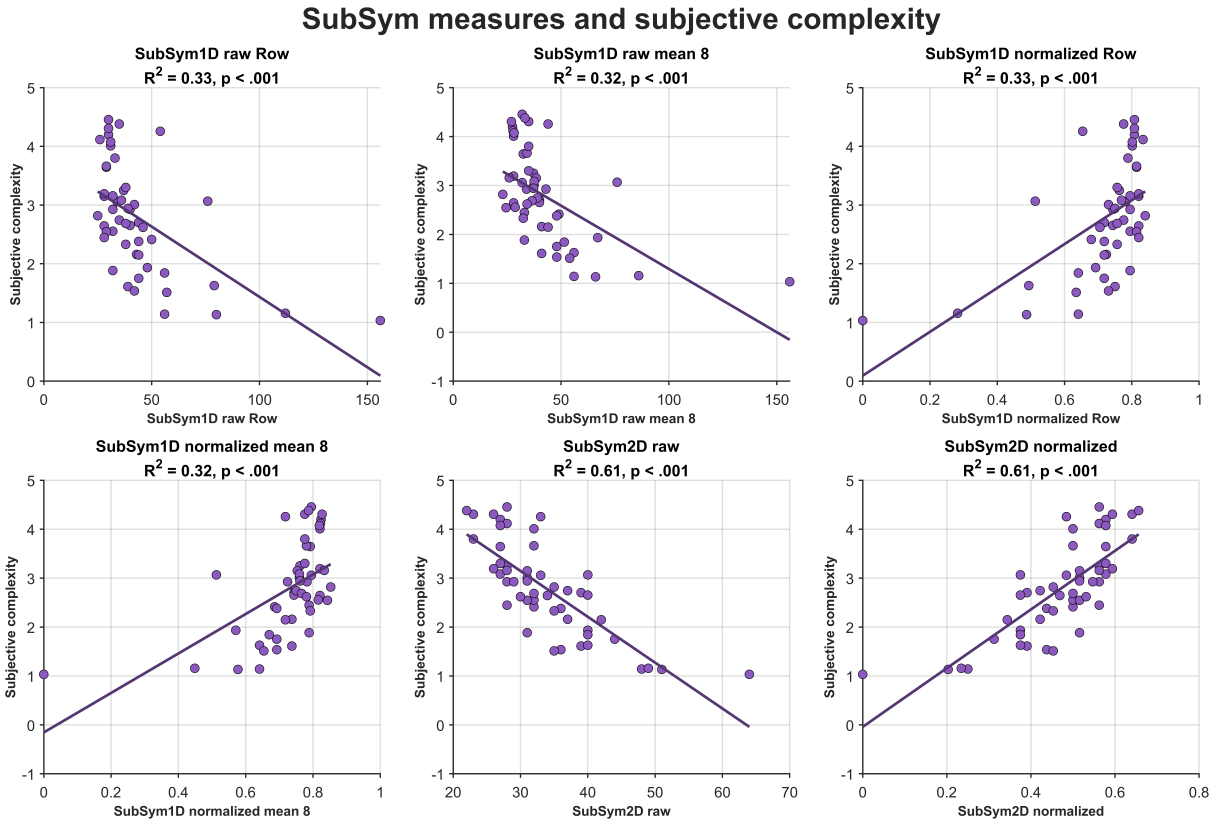


Figure 7. The SubSym proxy for subjective complexity. The panels show one-dimensional and two-dimensional SubSym indices. Each point represents one 3×3 binary pattern. Each panel includes the fitted linear trend and the corresponding R^2 and p value. These plots provide a structural-compression test of whether perceived complexity decreases when a pattern contains more symmetric substructure.

Discrete Structure of the Group-Theoretic Predictor

Recall that Garner's R&R set size is a discrete group-theoretic predictor. In the present stimulus set, $R(X)$ takes the ordered values 1, 2, 4, and 8. These values reflect increasing orbit sizes under the R &R of the square. The respective numbers of patterns at these four levels were 4, 4, 25, and 18. Again, the regression lines involving this variable speak to the order relations across these unbalanced orbit-size classes rather than to continuous relations. They should not be interpreted as evidence for continuous relations. This is not a problem of the R & R measure, but part of the theory, which predicts that good forms generate fewer distinct alternatives under transformation.

The mean figural-goodness ratings at $R(X) = 1$, 2, 4, and 8 were 5.90, 5.58, 4.14, and 2.33, respectively. The corresponding aesthetic-value means were 5.37, 5.04, 4.07, and 2.48, whereas the subjective-complexity means were 1.97, 1.91, 2.68, and 3.32. Thus,

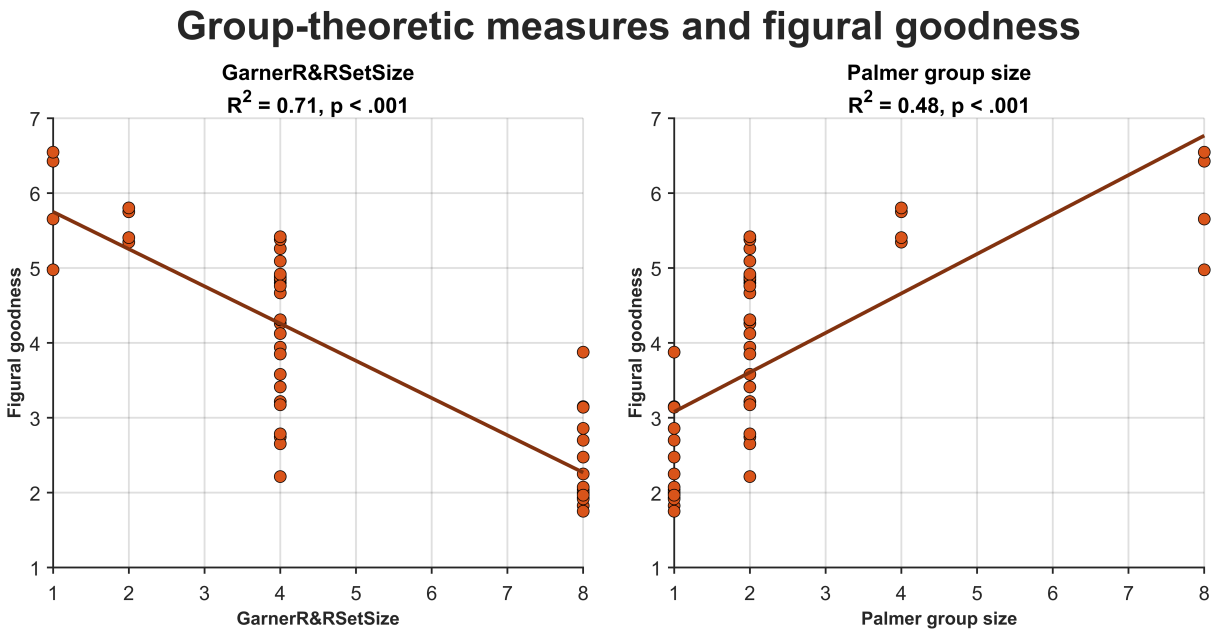


Figure 8. The Gestalt/group-theoretic proxy for figural goodness. The panels show Garner's rotation-and-reflection set size and Palmer's group size. Each point represents one 3×3 binary pattern. Each panel includes the fitted linear trend and the corresponding R^2 and p value. Because these predictors take only discrete group-theoretic values, the fitted lines are intended as visual summaries of ordered trends rather than as claims about continuously sampled predictors.

despite the unequal cell sizes, the ordered class means followed the predicted group-theoretic trend.

Reflection-Axis Check

We also tested whether the influence of the R &R variable could be explained by the presence of a mirror-reflection axis. Notably, we found that figural goodness could be predicted by the presence of any reflection axis, $R^2 = .50$, $p < .001$. However, this measure accounted for less variance than the full R &R measure, $R^2 = .71$, $p < .001$. Importantly, when we entered into the regression both predictors, the reflection indicator was not significant, $\beta = -.25$, $p = .164$, but the R &R measure remained strongly predictive, $\beta = -1.06$, $p < .001$. A model containing indicators for the four specific reflection axes accounted for $R^2 = .70$, but generalized less well than the R &R model, LOOCV $R^2 = .61$ versus .69. In the combined model, vertical reflection made an additional positive contribution, $\beta = .33$, $p = .004$, while the R &R measure remained significant, $\beta = -.67$, $p < .001$. This model yielded $R^2 = .79$ and LOOCV $R^2 = .71$. The

upshot is that the effect of R &R measure was not reducible to the mere presence of mirror symmetry, although vertical reflection provided additional information beyond orbit size.

Table 3 summarizes the performance of the primary regression models used to predict figural goodness. One may notice that the algorithmic and local-statistical variables exhibited negative LOOCV R^2 values. However, this finding should not be interpreted as showing that their individual predictors were necessarily uninformative. This can be attributed to the substantial redundancy among predictors and/or the relatively small number of stimulus patterns. These contributed to their instability.

Table 3
Model comparison for predicting figural goodness

Model	R^2	Adjusted R^2	LOOCV R^2	LOOCV RMSE
Main predictor model	.726	.696	.635	.598
Garner rotation-and-reflection set size	.705	.699	.685	.556
Palmer group size	.480	.470	.429	.748
Algorithmic predictors	.162	.069	−.215	1.091
Reduced local-statistical predictors	.152	.078	−.032	1.006

Note. LOOCV = leave-one-out cross-validation; RMSE = root mean squared error. The Garner-only model provided the best cross-validated performance, indicating that the Gestalt/group-theoretic proxy captured the central organizational property underlying figural-goodness judgments.

Algorithmic and local proxies do not account for figural goodness

The relations between local-statistical, algorithmic, and SubSym proxies and figural goodness were generally weak. Some one-dimensional compression-based measures showed modest negative relations with figural goodness, suggesting that more compressible patterns tended to be judged as somewhat better. However, these relations were much smaller than those observed for the group-theoretic proxy. Normalized two-dimensional SubSym complexity was related to figural goodness, $r(49) = -.35$, $p = .012$, $R^2 = .12$, but the effect was much smaller than the effect of Garner's R &R set size. Lempel–Ziv primitive complexity averaged across eight readings was also related to figural goodness, $r(49) = -.30$, $p = .036$, $R^2 = .09$, as was Effort-To-Compress averaged across eight readings, $r(49) = -.29$, $p = .040$, $R^2 = .08$. Zenil et al.'s two-dimensional

Coding Theorem Method estimate, $K_{m,2D}$, was essentially unrelated to figural goodness, $r(49) = -.04$, $p = .760$, $R^2 = .002$.

This result is theoretically important because it shows that figural goodness is not simply the inverse of complexity. If figural goodness were reducible to low algorithmic complexity, then Zenil complexity, Lempel–Ziv complexity, and Effort-To-Compress should have been among the strongest predictors of goodness. If figural goodness were reducible to local image statistics, then density, entropy, transitions, or fragmentation should have explained the judgments. Instead, the strongest predictor was the group-theoretic proxy for global organization. The near-zero relation between density and figural goodness, $r = -.06$, is especially informative because it argues against the possibility that Garner's measure merely serves as a proxy for the number of filled cells.

Aesthetic value is partly aligned with the Gestalt proxy

Aesthetic value showed a pattern similar to figural goodness, although the results were somewhat weaker. Figure 9 presents the group-theoretic comparison. Garner's rotation-and-reflection set size was strongly negatively related to aesthetic value, $r(49) = -.80$, $p < .001$, $R^2 = .64$, and Palmer's group-size measure was positively related to aesthetic value, $r(49) = .63$, $p < .001$, $R^2 = .40$. The same main-predictor model used for figural goodness also accounted for substantial variance in aesthetic value, $R^2 = .69$, adjusted $R^2 = .66$, with LOOCV $R^2 = .58$. Aesthetic value and figural goodness were themselves very strongly correlated, $r(49) = .96$, $p < .001$. Thus, aesthetic value provided a highly convergent judgment in this stimulus set rather than a clearly independent subjective dimension.

One issue that deserves comment concerns the very strong correlation between aesthetic value and figural goodness. The two construct may be dissociable, but in the current study they seem to be governed by the group-theoretic structure. The question of whether the two constructs dissociate should be tested with richer stimuli and additional determinants such as familiarity, interest, and individual preference.

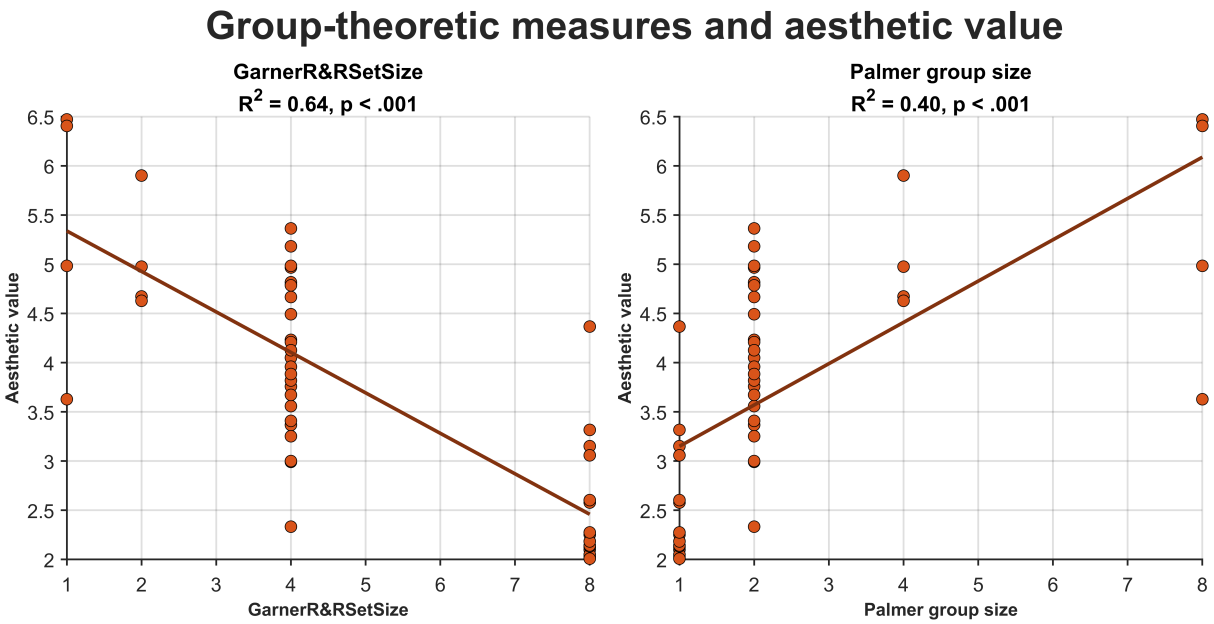


Figure 9. The Gestalt/group-theoretic proxy for aesthetic value. The panels show Garner's rotation-and-reflection set size and Palmer's group size. Each point represents one 3×3 binary pattern. Each panel includes the fitted linear trend and the corresponding R^2 and p value. The fitted lines serve as visual summaries of ordered trends.

Exploratory quadratic checks

The exploratory quadratic models provided no evidence that aesthetic value or figural goodness peaked at an intermediate level of any tested complexity measure. One uncorrected quadratic effect was obtained for subjective complexity, but it was one result among 21 exploratory tests and is not interpreted further.

Individual Differences in Subjective Judgments

To examine whether the stimulus-level relationships were also observed within individual participants, we calculated participant-level correlations across the 51 patterns. All 122 participants showed positive correlations between subjective complexity and both density and normalized two-dimensional SubSym. Approximately 97% showed negative correlations between Garner's rotation-and-reflection set size and both figural goodness and aesthetic value. The distributions of these individual correlations are reported in the Appendix.

Discussion

The present study compared several objective descriptions of 3×3 visual patterns, but the comparison was not intended as a purely numerical contest among measures. The measures were treated as operational proxies for competing theories of perception (Algom & Fitousi, 2025; Fitousi & Algom, 2024). Local-statistical measures operationalized a summary-statistic account (Attneave, 1954; Hochberg & McAlister, 1953; Shannon, 1948); algorithmic and compression-based measures operationalized a description-length or recoding account (Kolmogorov, 1965; Li & Vitányi, 2008; Zenil et al., 2015); and group-theoretic measures operationalized a Gestalt account of global organization (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). Our results suggest that subjective complexity is related mainly to local-statistical and compression-like measures, whereas figural goodness is related more strongly to Garner's and Palmer's group-theoretic measures (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). This distinction reinforces Garner's observation that good patterns remain invariant under more rotations and reflections.

This conclusion has direct implications for the study of figural goodness. A common assumption is that good figures are simply simple figures, and that complexity measures should therefore predict figural goodness by inverse relation (Attneave, 1954; Garner, 1962; Garner & Clement, 1963). The present framework suggests that this assumption is too broad. This is because it collapses distinct perceptual theories into a single complexity label (Algom & Fitousi, 2025; Fitousi & Algom, 2024). Some measures quantify local variation, some quantify compressibility, and some quantify global invariance. These properties can coincide, but they can also dissociate. A pattern may be locally heterogeneous but globally symmetric, while another pattern may be locally homogeneous but visually uninteresting. Figural goodness appears to depend especially on the organizational structure of the pattern rather than on complexity in the abstract (Garner, 1970; Palmer, 1991).

Our additional analysis of the contribution of reflection-axis strengthens this interpretation. Garner's orbit-size measure remained strongly predictive after the

presence of mirror reflection was controlled. This supports the notion that its effect was not reducible to a binary mirror-symmetry distinction. However, vertical reflection explained additional variance beyond Garner's measure. The results therefore support the broader group-theoretic account. But they also align with the evidence that vertical reflection has a privileged perceptual status (Bertamini et al., 2018; Wagemans, 1997; Wagemans, Van Gool, Swinnen, & Van Horebeek, 1993).

An alternative account of perceptual goodness is the holographic model of van der Helm and Leeuwenberg (1996). This account evaluates the strength of evidence for visual regularity relative to the total information in the pattern. It expresses it as $W = E/N$. Here, E represents the number of holographic identities supporting a regularity, and N represents the total number of elements. The holographic and transformational accounts can therefore make different predictions about the relative goodness of patterns with different symmetries (Wenderoth & Welsh, 1998). The relevance of global and local symmetries to regularity detection has also been demonstrated empirically (Nucci & Wagemans, 2007). Further evidence links visual regularity to electrophysiological responses (Makin et al., 2016) and aesthetic preference (Makin, Helmy, & Bertamini, 2018). Perspective slant can impair symmetry detection and reduce aesthetic appeal (Bertamini, Tyson-Carr, & Makin, 2022). Although we did not calculate W for the present patterns, the holographic model provides an important alternative account of the relationship between visual organization and figural goodness.

The distinction between algorithmic complexity and perceptual organization is especially important. Algorithmic complexity provides a principled mathematical account of description length and generative probability (Chaitin, 1969; Kolmogorov, 1965; Levin, 1974; Li & Vitányi, 2008; Solomonoff, 1964). It is therefore a powerful tool for classifying formal structure. However, human observers do not necessarily perceive patterns as universal Turing machines. They are visual systems with sensitivity to axes, balance, grouping, closure, and symmetry (Palmer, 1991; Pomerantz & Kubovy, 1986; Wagemans, Elder, et al., 2012; Wagemans, Feldman, et al., 2012). Consequently, a complete account of figural goodness must include computational measures that are psychologically aligned

with visual organization (Algom & Fitousi, 2025; Fitousi & Algom, 2024).

Interestingly, Zenil's $K_{m,2D}$ estimates showed moderate correlations with other objective measures of complexity, including SubSym, Lempel–Ziv, and Effort-to-Compress (rs ranging from .39 to .49). However, the association between Zenil's estimates and subjective complexity was relatively weak ($r = .26$, $p = .068$). This suggests that convergence among objective measures of complexity does not necessarily imply comparable psychological validity.

Mental Programs and Domain-Specific Complexity

Dehaene et al. (2022) have proposed mental-program and language-of-thought accounts in which humans represent structured stimuli using compact descriptions based on domain-specific primitives, including repetition, rotation, reflection, and recursion (Amalric et al., 2017; Planton et al., 2021; Sablé-Meyer et al., 2022). For example, Amalric et al. (2017) found that the minimal description length of spatial sequences in a geometric language predicted performance better than distances between successive locations or a language lacking explicit symmetry primitives.

Our current findings suggest a similar conclusion for static patterns. The contrast between generic complexity and group-theoretic measures suggests that good form may not simply depend on a short description in an arbitrary computational language, but rather a short description in a geometrical language containing perceptually salient transformations (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). This does not mean that Garner's and Palmer's measures are equivalent to Kolmogorov complexity (Kolmogorov, 1965; Li & Vitányi, 2008). These are not measures of the shortest program that generates the individual cells of a pattern. The conclusion is that the relevant form of algorithmic complexity may be domain-specific. That is, figural goodness depends on compact descriptions constructed from perceptually meaningful geometrical operations (Amalric et al., 2017; Dehaene et al., 2022)

Transformational Organization and Aesthetic Value

Although the central theoretical contrast concerned subjective complexity and figural goodness, aesthetic value provided an important converging outcome. Aesthetic value was strongly related to figural goodness, $r(49) = .96$, $p < .001$, indicating that the two judgments were closely aligned for the present stimulus set. Garner's rotation-and-reflection set size was also strongly related to aesthetic value, although its cross-validated fit was somewhat lower for aesthetic value than for figural goodness.

The somewhat lower cross-validated fit for aesthetic value may indicate that aesthetic evaluation also depends on factors not represented by the present structural measures, including familiarity, learning history, and individual preference. However, because aesthetic value and figural goodness were so strongly related at the pattern level, the present results do not establish that they represent clearly distinct psychological constructs.

Theoretical Implications

Our approach by which theories of perception are tested via families of measures is constructive. It can explain previous inconsistent conclusions about the role of complexity in preference and goodness (Algom & Fitousi, 2025; Berlyne, 1971; Birkhoff, 1933; Fitousi & Algom, 2024). It is likely that such measures as entropy, Lempel–Ziv complexity, CTM complexity, SubSym, and group-theoretic invariance are not measuring the same construct. These measures reflect different assumptions about the primitives and operations that determine complexity (Lempel & Ziv, 1976; Nagaraj & Balasubramanian, 2017; Palmer, 1991; Shannon, 1948; Zenil et al., 2015; Ziv & Lempel, 1977). A major contribution of the present study is in testing all of them directly with a single set of stimuli and participants.

The present study considers two-dimensional patterns presented in a fixed orientation. However, in natural viewing conditions, the retinal representation of a pattern may change considerably with viewpoint, and the symmetries of the distal stimulus may not be preserved in the retinal image. Indeed, Bertamini et al. (2022)

showed that perspective distortions impair symmetry detection and reduce aesthetic preference. From a group-theoretic perspective, an important distinction should therefore be made between the transformations that leave a pattern invariant and the transformations induced by changes in viewpoint. Although distal symmetry does not guarantee invariance under perspective projection, it may facilitate the recovery of stable representations across viewpoints. The present findings do not distinguish between distal and retinal determinants of perceptual goodness, but suggest that their relationship may be understood in terms of symmetry and invariance. Examining how the structural measures considered here predict judgments across changes in viewpoint would be an interesting direction for future research.

Limitations

Several limitations should be noted. First, the present stimuli were very small binary patterns. This restriction allowed exhaustive comparison across a well-defined stimulus space, but it remains unclear how the same measures scale to larger and more naturalistic images. Second, some measures require reducing a two-dimensional pattern to a one-dimensional string. Such reductions may lose spatial information that is psychologically important. Third, each participant evaluated every pattern on all three judgment dimensions. Although task order and pattern order were randomized, prior exposure to a pattern in one task may have influenced its subsequent evaluation in another task. Because the order of the judgment tasks was not retained in the available data, we could not examine these potential carryover effects retrospectively. Future studies could assign the judgment dimensions to independent groups of participants to eliminate such effects. Fourth, the analyses were correlational and predictive. They identify which theoretical proxies account for the judgments, but they do not reveal the processing mechanisms through which observers derive those judgments. In particular, the present study does not demonstrate that observers explicitly execute rotations, reflections, or other mental-program operations. Process-level predictions involving scrambling, presentation duration, working-memory load, transformation priming, and

sequential recoding remain important targets for future experiments. Another limitation concerns the formulation of the subjective rating scales. Although all three judgments were made on 7-point scales, their verbal anchors were not formulated consistently. We cannot rule out the possibility that these differences influenced participants' interpretations of the scales and, consequently, their judgments.

Conclusion

The study of visual complexity has progressed from qualitative Gestalt principles to formal information-theoretic and algorithmic measures (Attneave, 1954; Kolmogorov, 1965; Li & Vitányi, 2008; Shannon, 1948; Wagemans, Elder, et al., 2012; Wagemans, Feldman, et al., 2012). The present paper argues that the next step is not to select one measure as the true measure of complexity, but to ask which theory of perception each measure operationalizes (Algom & Fitousi, 2025, 2026; Fitousi & Algom, 2024). Local statistical measures serve as proxies for summary-statistic accounts, algorithmic measures serve as proxies for description-length and recoding accounts, and Gestalt measures serve as proxies for global organization and transformation invariance (Garner, 1970; Garner & Clement, 1963; Palmer, 1991). Human judgments of figural goodness are most naturally explained by this theoretical family structure. Viewed through the lens of mental-program accounts, the critical issue is not whether a pattern can be compressed in the abstract, but whether it can be compressed by geometrically meaningful operations such as symmetry, repetition with variation, and transformational invariance (Dehaene et al., 2022; Sablé-Meyer et al., 2022). Good figures are not merely low-complexity figures; they are organized figures.

Appendix: Individual Differences in Subjective Judgments

To assess whether the stimulus-level relationships were also observed within individual participants, we calculated participant-level Pearson correlations between selected objective measures and ratings across the 51 patterns. The analyses used 122 participant rows. Invalid or missing ratings were omitted on a per-correlation basis.

Participant-level relationships with objective measures. For each participant, we calculated Pearson correlations across the 51 patterns between the values of a given objective measure and the participant's corresponding subjective ratings. This procedure was performed separately for four pairs of objective measures and subjective judgments: (1) density and subjective complexity, (2) normalized two-dimensional SubSym and subjective complexity, (3) Garner's rotation-and-reflection set size and figural goodness, and (4) Garner's rotation-and-reflection set size and aesthetic value. Thus, for each pair, we obtained a distribution of 122 individual correlation coefficients. Table A1 summarizes the resulting distributions of signed correlations. Squared correlations are also included to facilitate comparison with the stimulus-level R^2 values, although the sign of the correlation is important for interpreting agreement across participants.

Table A1
Participant-level correlations between structural measures and subjective ratings

Comparison	Median r	r IQR	Median R^2	Expected sign (%)
Density–complexity	.569	[.445, .670]	.324	100.0
SubSym 2D–complexity	.575	[.502, .652]	.331	100.0
Garner–goodness	−.657	[−.741, −.507]	.432	96.7
Garner–aesthetic value	−.597	[−.671, −.438]	.357	96.7

Note. Each comparison was based on 122 valid participant correlations. IQR = first and third quartiles of individual signed correlations. Expected sign indicates the percentage of participants whose correlation had the direction predicted by the corresponding account.

Figure A1 shows the distributions of the individual Pearson correlations for the four comparisons. Each panel shows the number of participants at each correlation level, the zero reference line, and the median individual correlation.

The direction of association was consistent across participants for the two complexity comparisons. For Garner's measure, 96.7% of participants showed the

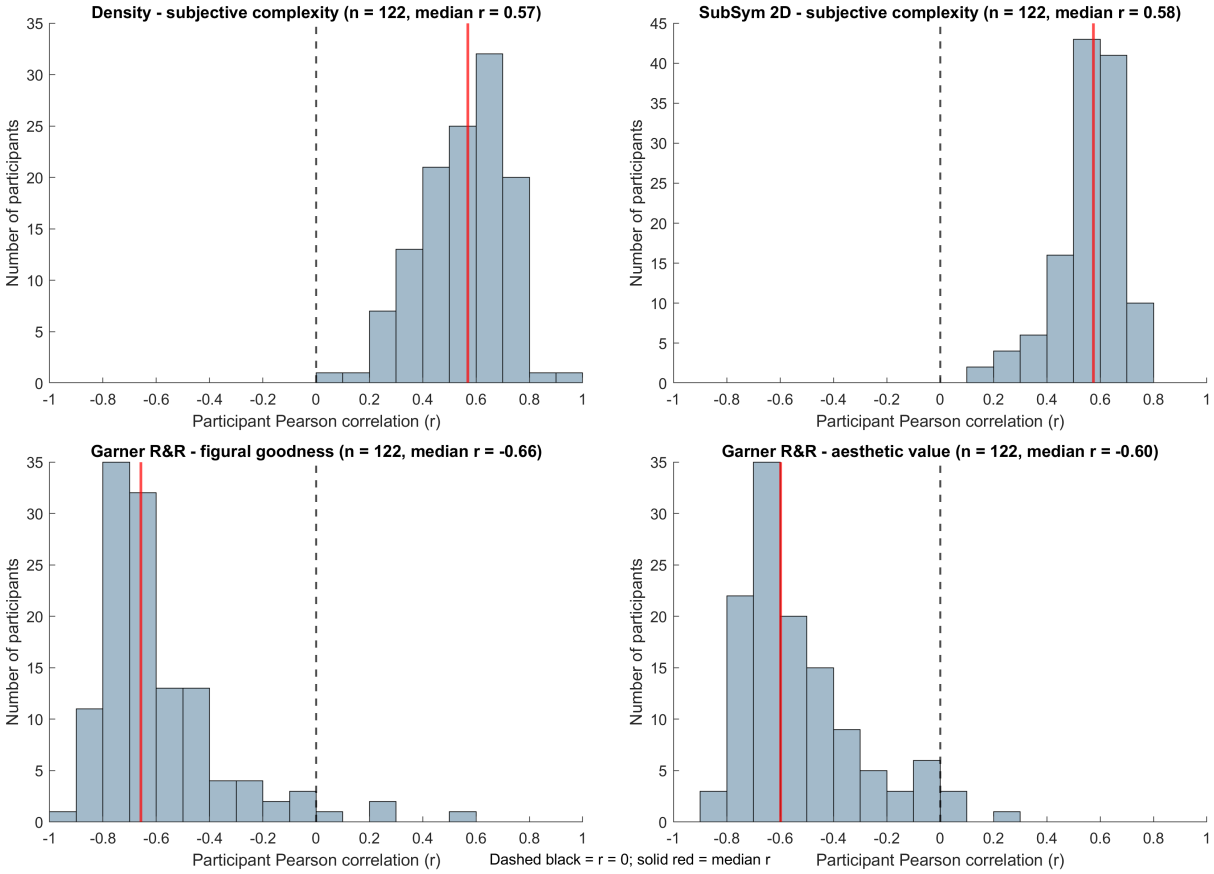


Figure A1. Distributions of participant-level Pearson correlations across 51 patterns for density and subjective complexity (upper left), normalized two-dimensional SubSym and subjective complexity (upper right), Garner's rotation-and-reflection set size and figural goodness (lower left), and Garner's set size and aesthetic value (lower right). Dashed black lines indicate zero correlation; solid red lines indicate the median correlation. Each panel includes 122 participants.

predicted negative association with each evaluative judgment. These results indicate that the stimulus-level associations were not solely the product of averaging across observers with opposing trends.

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